

Booleans & Conditionals

BILD 62

By the end of
this lecture you
will be able to:

- Recognize Booleans & write conditional logic statements
 - Test conditional statements in Python
 - Demonstrate you can trace code
-

Operators in Python

Operators are special symbols that carry out arithmetic or logical computation.

Type of operator	Examples
assignment	<code>a = 6</code>
arithmetic (math)	<code>2 * 3</code>
logic (boolean)	<code>True and False</code>
comparison (<i>more on next slide</i>)	<code>a != 6</code>
identity	<code>a is 6</code>
membership	<code>'a' in 'cat'</code>

Basic comparative operators in Python

Symbol	Operation	Usage	Outcome
<code>==</code>	Is equal to	<code>10==5*2</code>	True
<code>!=</code>	Is not equal to	<code>10 != 5*2</code>	False
<code>></code>	Is greater than	<code>10 > 2</code>	True
<code><</code>	Is less than	<code>10 < 2</code>	False
<code>>=</code>	Greater than <i>or</i> equal to	<code>10 >= 10</code>	True
<code><=</code>	Less than <i>or</i> equal to	<code>10 <= 10</code>	True

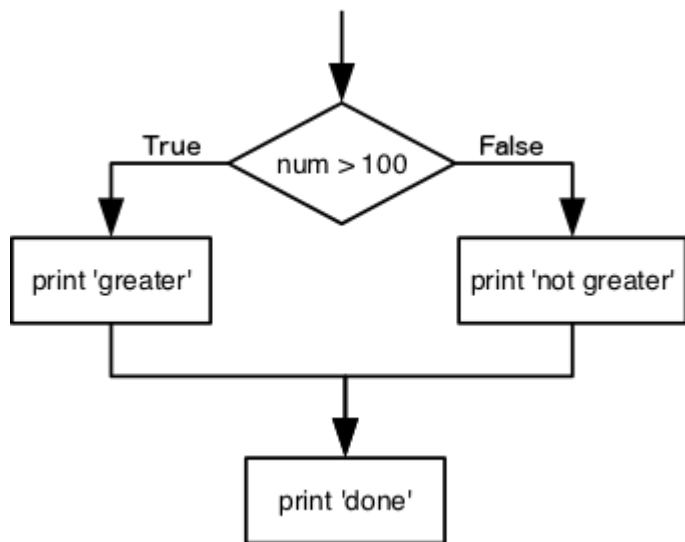
**Boolean variables
store `True` (1) or
`False` (0) and are
the basis of all
computer
operations.**

Sydney Padua:

<https://sydneyadua.com/2dgoggles/happy-200th-birthday-george-boole/>



```
num = 37
if num > 100:
    print('greater')
else:
    print('not greater')
print('done')
```



Conditional statements

help us make decisions
in our code.

(We can also call it
control flow.)

if statements syntax

`if` condition:  you need a colon here!

indented
by 4 spaces
(or tab)

```
    print('condition met')  
    print('nice work.')  
print('not in the block')
```

 block

if/else statement syntax

if condition:

```
print('condition met')
```

```
print('nice work.')
```

else:

```
print('condition not met')
```



you need a
colon here!

One more conditional: **elif**

- Short for “else if”
- Enables you to check for additional conditions → *necessary if there is more than two outcomes*

```
condition_1 = False
condition_2 = True
```

```
if condition_1:
    print('Condition 1 is true.')
elif condition_2:
    print('Condition 2 is true.')
else:
    print('Both Condition 1 and 2 are false.')
```

Resources

[Intro. to Comp. Sci. & OOP: Python · Cogniterra](#)

[Plotting and Programming in Python: For Loops](#)

[Plotting and Programming in Python: Conditionals](#)

[Whirlwind Tour of Python: Control Flow](#)